

SWAPNIL SHARAD MERGU

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building Python-based backend services, RESTful APIs, web applications, and cloud-supported platforms across banking and SaaS environments. Hands-on with Django, FastAPI, Flask, React.js, MySQL, MongoDB, AWS, Docker, Pytest, and CI/CD, with contributions spanning API development, authentication, database optimization, testing, deployment, and production support. Improved API response time by up to 25%, reduced deployment effort by 30%, and strengthened release quality through automated testing and structured debugging. Brings practical experience in secure application development, Agile delivery, production support, and cross-functional collaboration across U.S. banking and SaaS product environments.

TECHNICAL SKILLS

- Programming Languages:** Python, Java, SQL, C++
- Backend Development:** Django, Django REST Framework, Flask, FastAPI, RESTful APIs, Microservices, API Integration, JWT Authentication
- Frontend Development:** React.js, React Native, JavaScript, HTML5, CSS3, Responsive Web Development
- Cloud & Platform Technologies:** AWS Lambda, Amazon EC2, Amazon S3, Amazon API Gateway, Amazon Cognito, AWS CloudWatch
- Databases:** MySQL, MongoDB, MongoDB Atlas, Amazon DynamoDB, Database Design, Query Optimization
- DevOps & Developer Tools:** Docker, Kubernetes, Git, GitHub, CI/CD, Linux, Postman, Swagger/OpenAPI, Visual Studio Code
- Software Engineering:** Object-Oriented Programming, Data Structures and Algorithms, Design Patterns, System Design Fundamentals, Distributed Systems, Performance Optimization
- Testing & Methodologies:** Pytest, Unit Testing, Integration Testing, API Testing, Test-Driven Development, Agile Scrum, Debugging, Code Reviews, SDLC
- AI & Emerging Technologies:** LangChain, Ollama, Mistral LLM, Large Language Models, Prompt Engineering, Natural Language Processing, Scikit-learn

PROFESSIONAL EXPERIENCE

Software Engineer **Dec 2025 - Present**
Truist Bank | Remote

- Developed Python-based RESTful APIs using FastAPI and Django REST Framework for account and transaction workflows, improving request processing speed by 18% across 6 application services.
- Modernized tightly coupled backend modules into reusable microservice components, reducing feature-development effort by 20% and simplifying maintenance across banking applications.
- Optimized SQL queries, joins, and database access patterns in MySQL, lowering response latency by 22% for customer and transaction-related functions.
- Secured application endpoints through JWT authentication, role-based access control, input validation, and structured error handling, reducing access-related defects by 30%.
- Introduced an AI-assisted support feature using Python, LangChain, Mistral LLM, and NLP to summarize errors and retrieve troubleshooting guidance, reducing incident analysis time by 21%.
- Automated unit, integration, and API testing using Pytest and Postman, increasing regression coverage to 85% and reducing release-validation defects by 24%.
- Containerized backend services with Docker and supported GitHub-based CI/CD workflows, decreasing environment-related deployment failures by 25%.
- Resolved production and lower-environment issues through CloudWatch logs, SQL validation, API tracing, and collaboration with QA, platform, and business teams, closing 90% of assigned incidents within service targets.

Software Engineer **May 2022 - May 2024**
Augur Talentcare Pvt Ltd | India | Remote

Client Domain: SaaS / Startup Product

Project: Cloud-Based Workflow and Customer Operations Platform

- Engineered Django and Flask backend services for customer onboarding, task management, user administration, and reporting, supporting more than 1,500 application records per day.
- Built RESTful APIs for React.js interfaces using Python, JSON, and Django REST Framework, reducing frontend data-processing errors by 23% across 8 product modules.
- Designed responsive web components with React.js, JavaScript, HTML5, and CSS3, lowering UI-related support issues by 18% across desktop and mobile users.

- Structured MySQL and MongoDB schemas, indexes, and CRUD operations for user, workflow, and configuration data, improving query execution time by 25%.
- Implemented JWT authentication and role-based permissions for administrators, managers, and standard users, reducing access-control defects by 28%.
- Refactored Python services using object-oriented design, reusable functions, caching, and asynchronous processing, decreasing API response time by 25%.
- Deployed containerized application components through Docker, AWS Lambda, and Amazon EC2, reducing manual deployment effort by 30% across development and testing environments.
- Collaborated with product managers, QA engineers, frontend developers, and support teams through Agile Scrum, completing 95% of committed sprint work across 10 release cycles.

PROJECTS

AI-Assisted Banking Application Support System

- Built an internal AI-assisted support application using Python, FastAPI, LangChain, and Mistral LLM to interpret application errors and retrieve relevant troubleshooting steps, reducing initial issue-analysis time by 23%.
- Connected MySQL-based incident records and approved support documentation through controlled natural-language queries, improving guidance accuracy by 19% across 6 recurring application issue categories.
- Delivered a React.js interface with prompt validation, response filtering, and human review controls, cutting repeated support searches by 26% without exposing customer or transaction data to the language model.

Multi-Tenant SaaS Configuration and Access Portal

- Created a multi-tenant portal with Django and React.js to manage customer configurations, user roles, and product settings, reducing administrative setup time by 30% for 40+ client accounts.
- Designed MongoDB collections and RESTful APIs for tenant-specific preferences, permissions, and workflow rules, improving configuration retrieval speed by 26% across 7 application modules.
- Crafted responsive React components using JavaScript, HTML5, and CSS3 for administrators and customer users, lowering configuration-related support requests by 19% after rollout.

Application Release Quality and Deployment Tracking Platform

- Developed a Flask-based release tracker that consolidated build status, test results, deployment notes, and defect ownership, improving release visibility across 3 engineering teams by 35%.
- Connected GitHub and CI/CD outputs through REST APIs to capture deployment outcomes and failed validation stages, reducing manual release-status reporting by 32% per sprint.
- Standardized API checks with Postman and Swagger/OpenAPI on Linux, cutting troubleshooting time by 21% and improving developer-QA handoffs.

EDUCATION

Master of Science in Computer Science

University of Central Missouri | Warrensburg, MO

May 2026

CERTIFICATIONS

- IBM DevOps and Software Engineering Professional Certificate - **Coursera**
- Meta Back-End Developer Professional Certificate - **Coursera**
- IBM Full Stack Software Developer Professional Certificate - **Coursera**
- AWS Cloud Technical Essentials - **Coursera**
- Software Development Lifecycle Specialization - **Coursera**
- Meta Front-End Developer Professional Certificate - **Coursera**